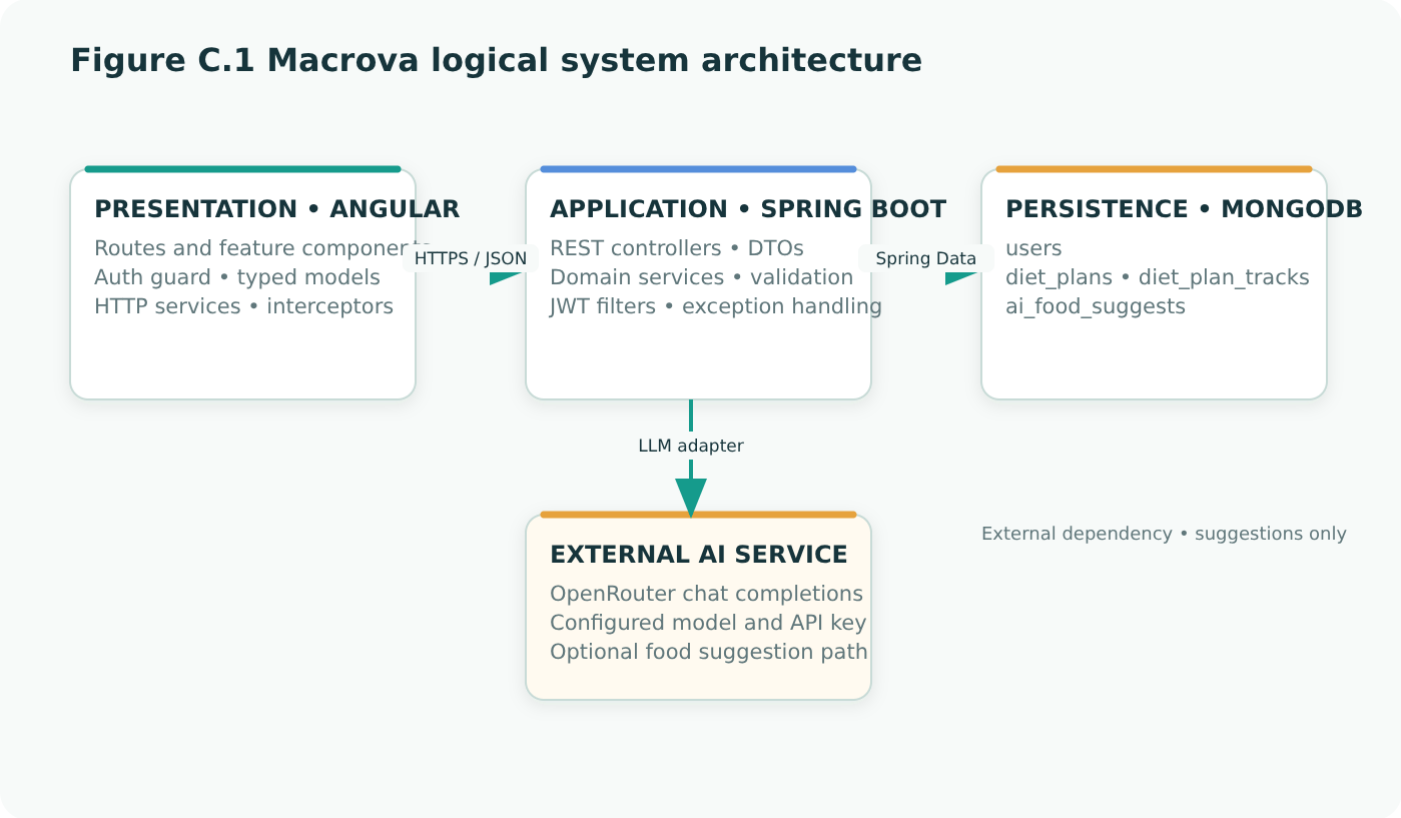
APPENDIX C — DIAGRAMS AND APPLICATION SCREENSHOTS

This visual appendix contains five system-design diagrams and six screenshots captured from the current Macrova Angular frontend. Diagrams are original illustrations based on the checked-in implementation. The authenticated UI examples use synthetic, mocked API responses for display only; no real account, production database, or external-model output is represented.

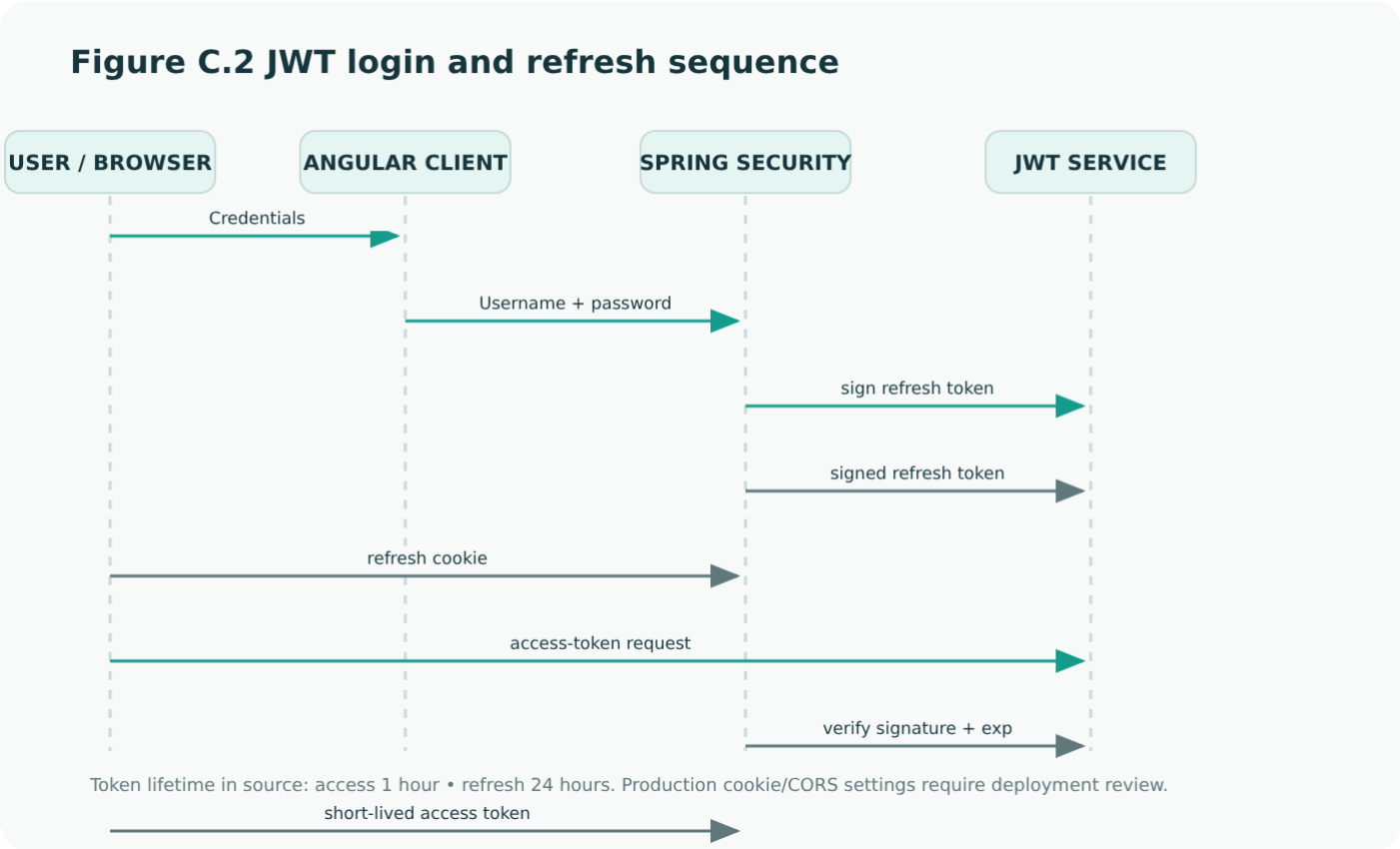
For a clearer view, the screenshots are cropped around the relevant interface content. Figures C.1–C.5 cover architecture, authentication, data, calculations, and AI flow; screenshots C.1–C.6 cover registration, sign-in, dashboard, daily logging, food suggestions, and plan creation.

Figure C.1 — Macrova logical system architecture



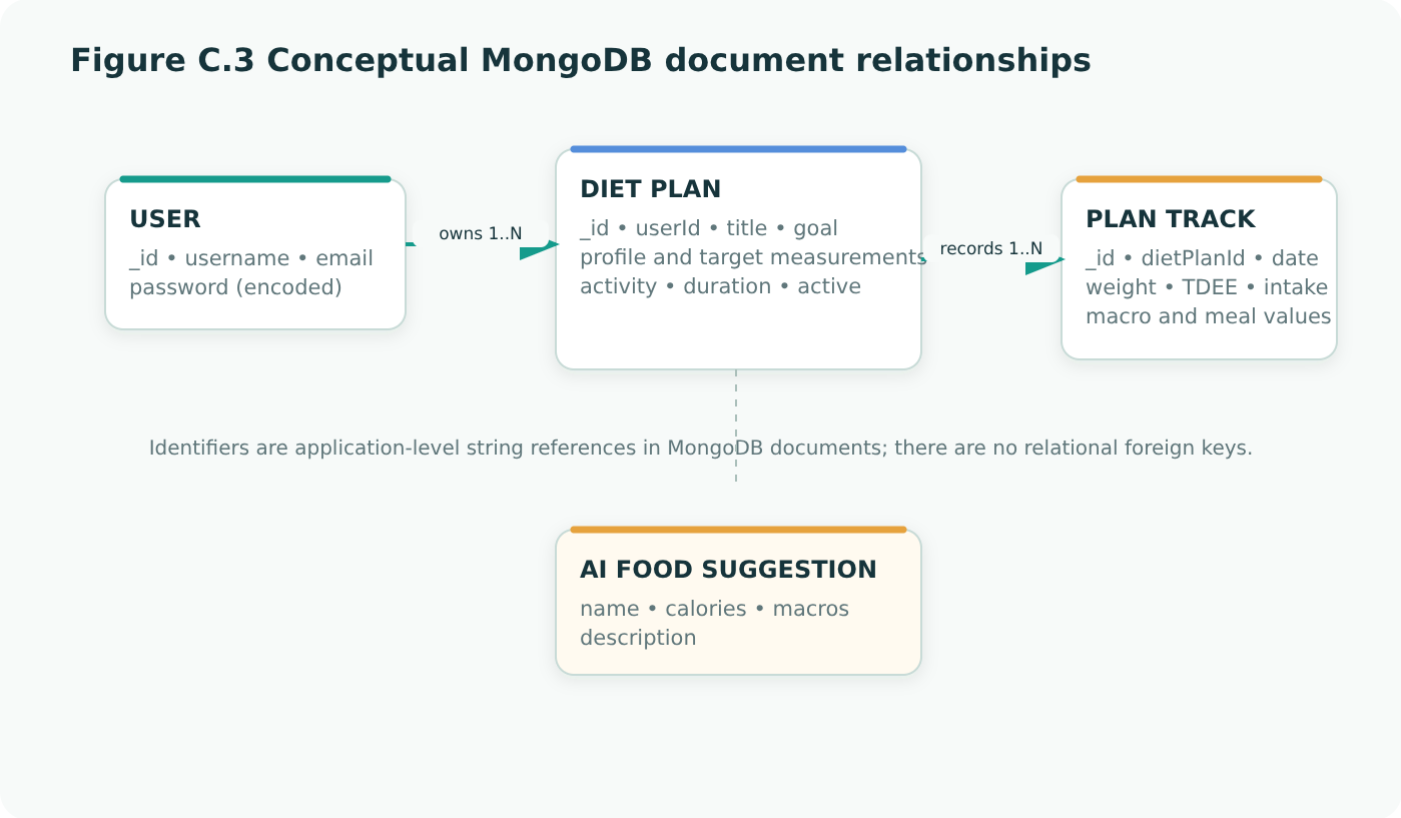
The diagram shows the Angular browser client, Spring Boot service layer, MongoDB persistence, and the external OpenRouter dependency used for meal suggestions.

Figure C.2 — JWT login and refresh sequence



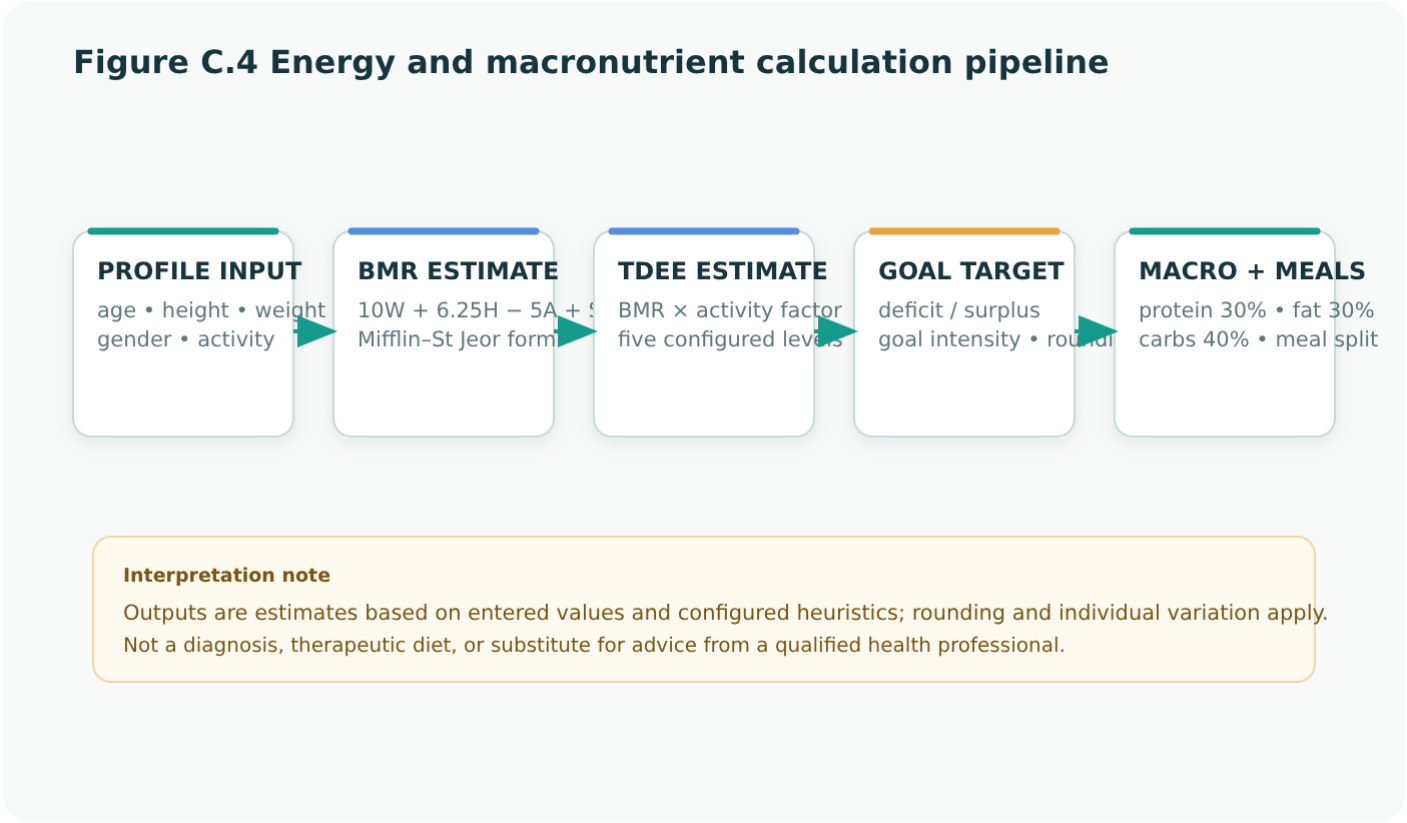
The sequence summarizes credential authentication, the refresh-cookie, and the access-token lifecycle present in the source. Production cookie and CORS settings require deployment review.

Figure C.3 — Conceptual MongoDB document relationships



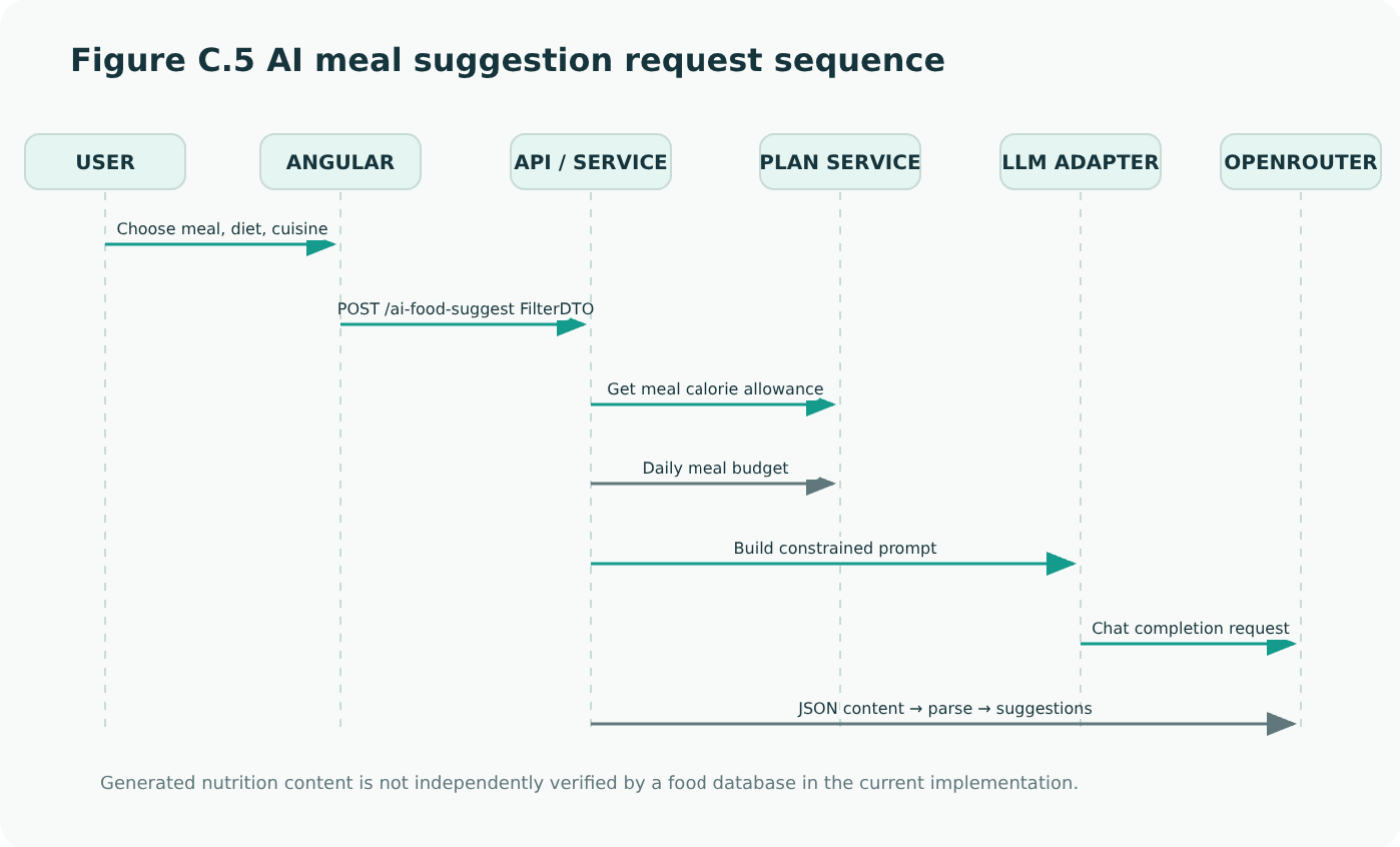
User, diet-plan, tracking, and food-suggestion documents are stored in separate MongoDB collections; identifier links are application-level references.

Figure C.4 — Energy and macronutrient calculation pipeline



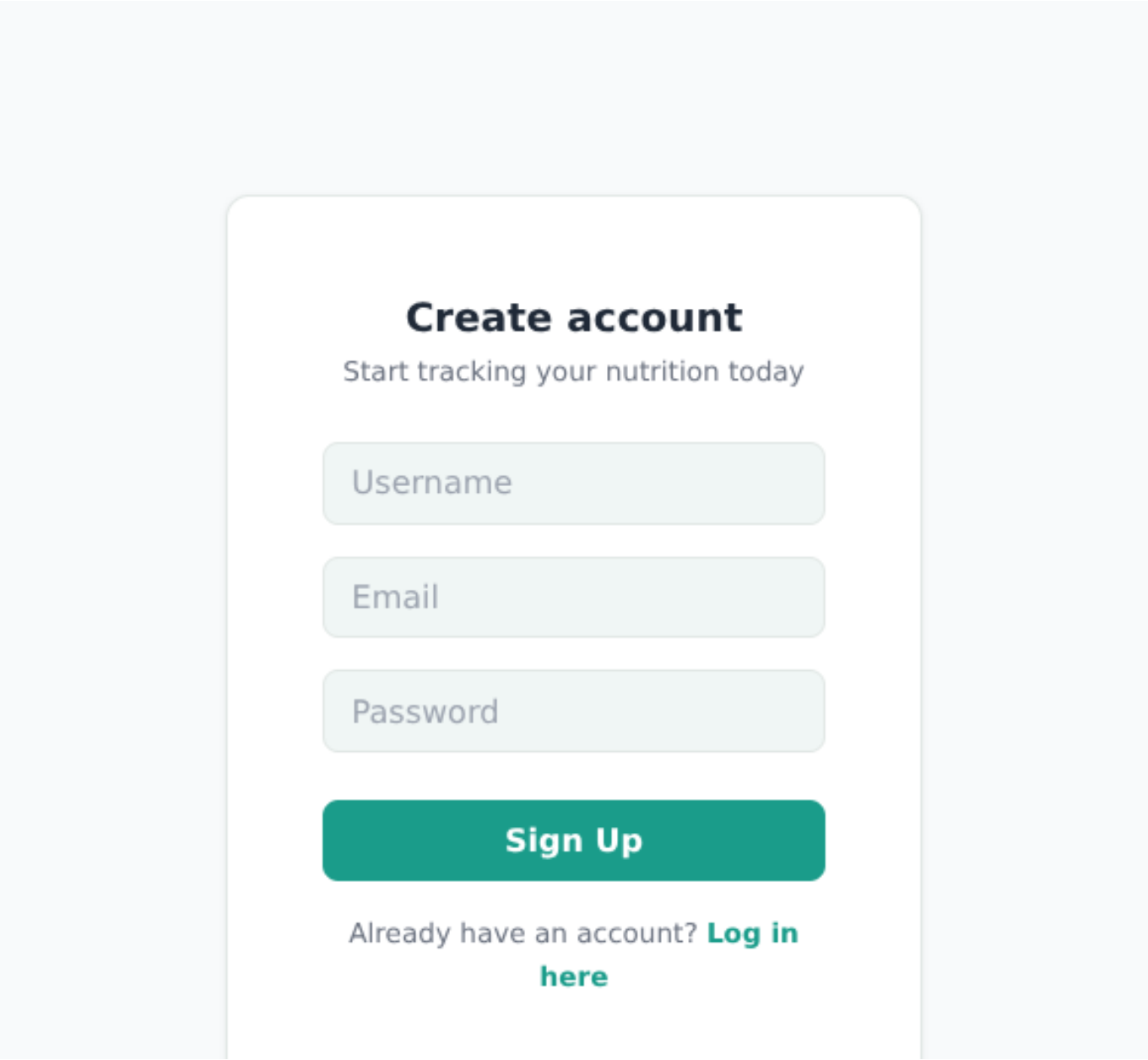
The flow summarizes the coded BMR/TDEE, goal adjustment, macro allocation, and meal split. Estimates are educational and not medical advice.

Figure C.5 — AI meal suggestion request sequence



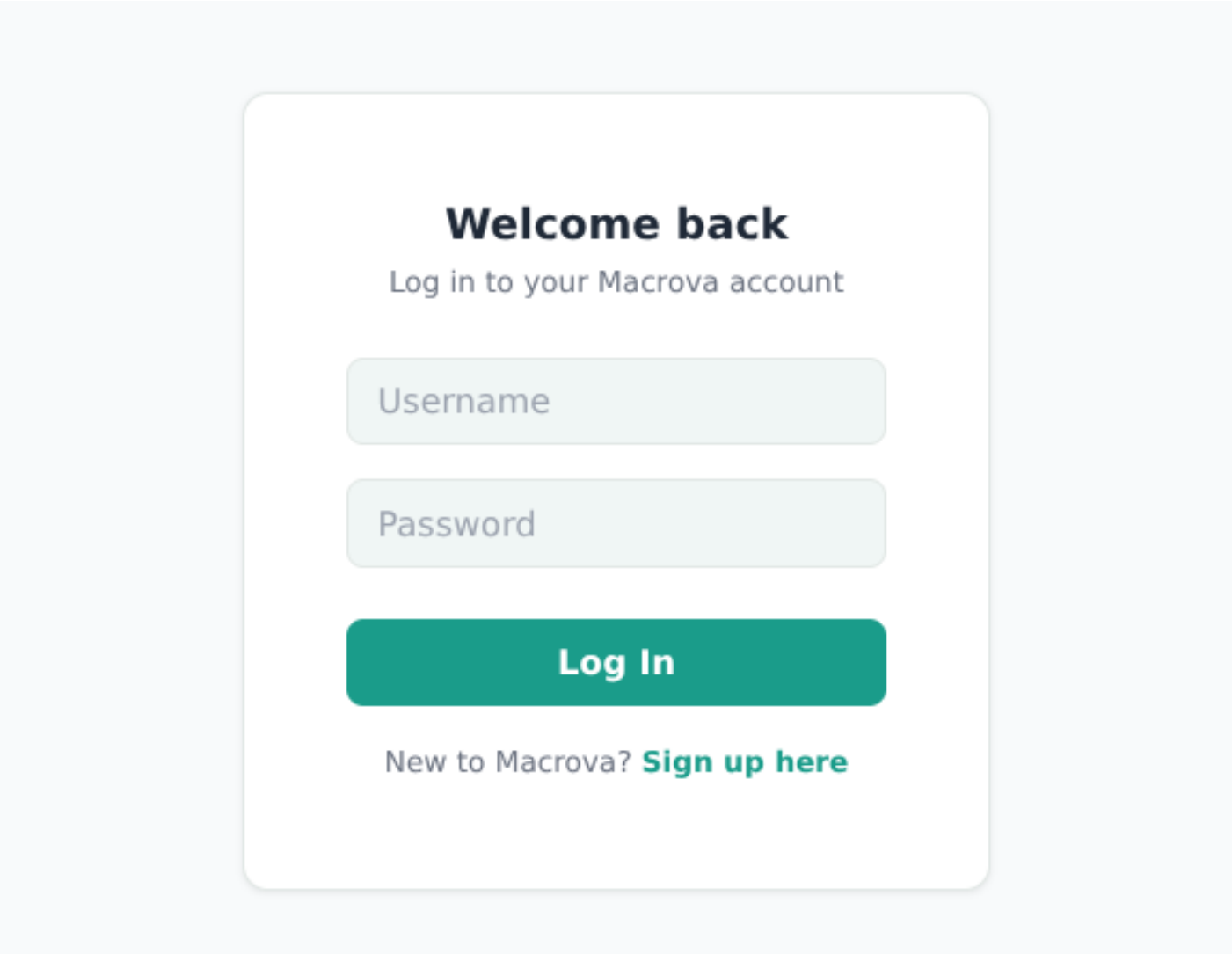
The flow depicts filter submission, meal-calorie lookup, prompt construction, OpenRouter call, and JSON parsing. Generated nutrition is not independently verified against a food database in the current code.

Screenshot C.1 — Macrova account registration



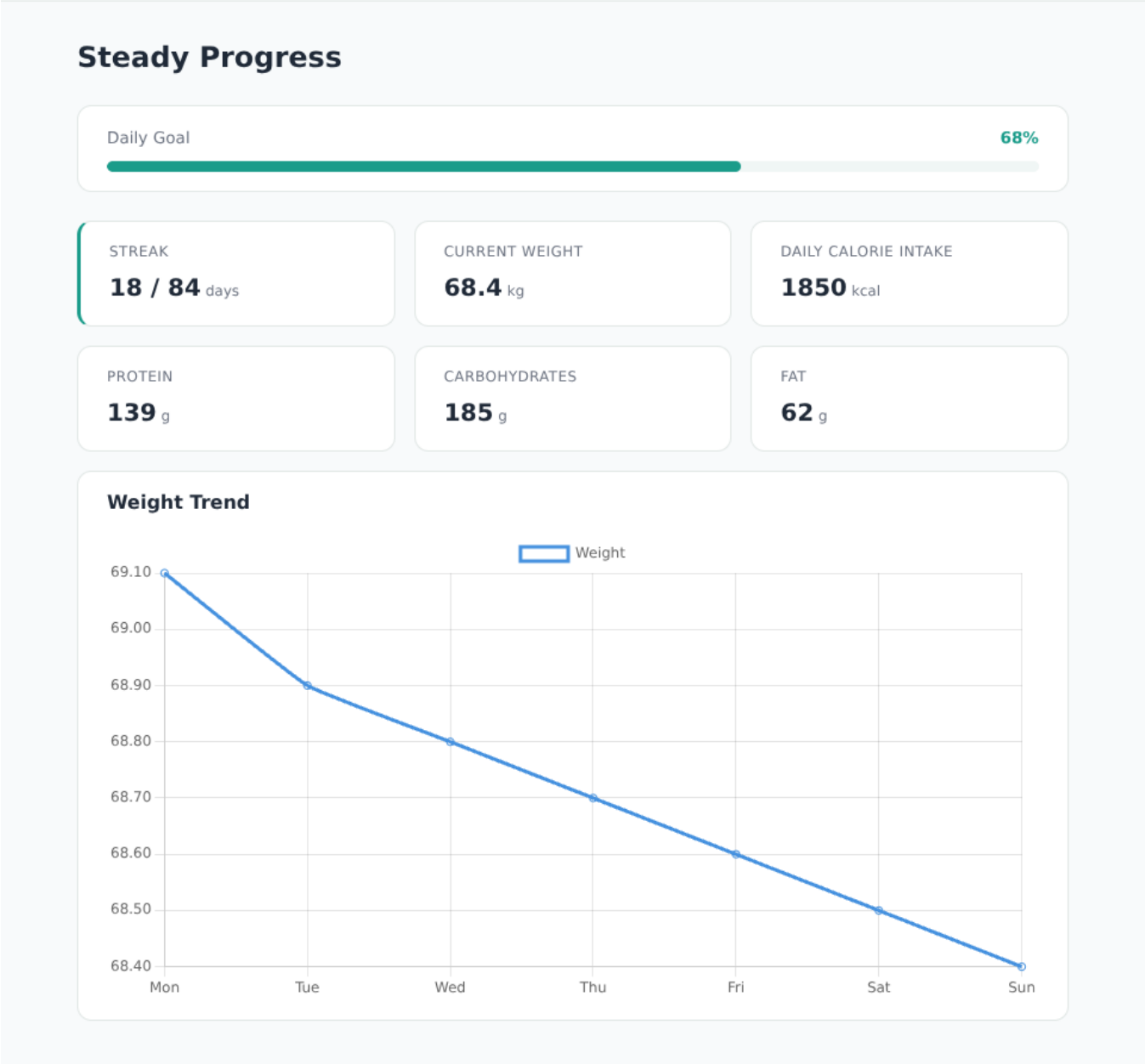
Captured from the running Angular application at /signup. This close crop highlights the actual registration component; no real account information is shown.

Screenshot C.2 — Macrova sign-in



Captured from the running Angular application at /login. This close crop highlights the actual sign-in component; no credentials were entered.

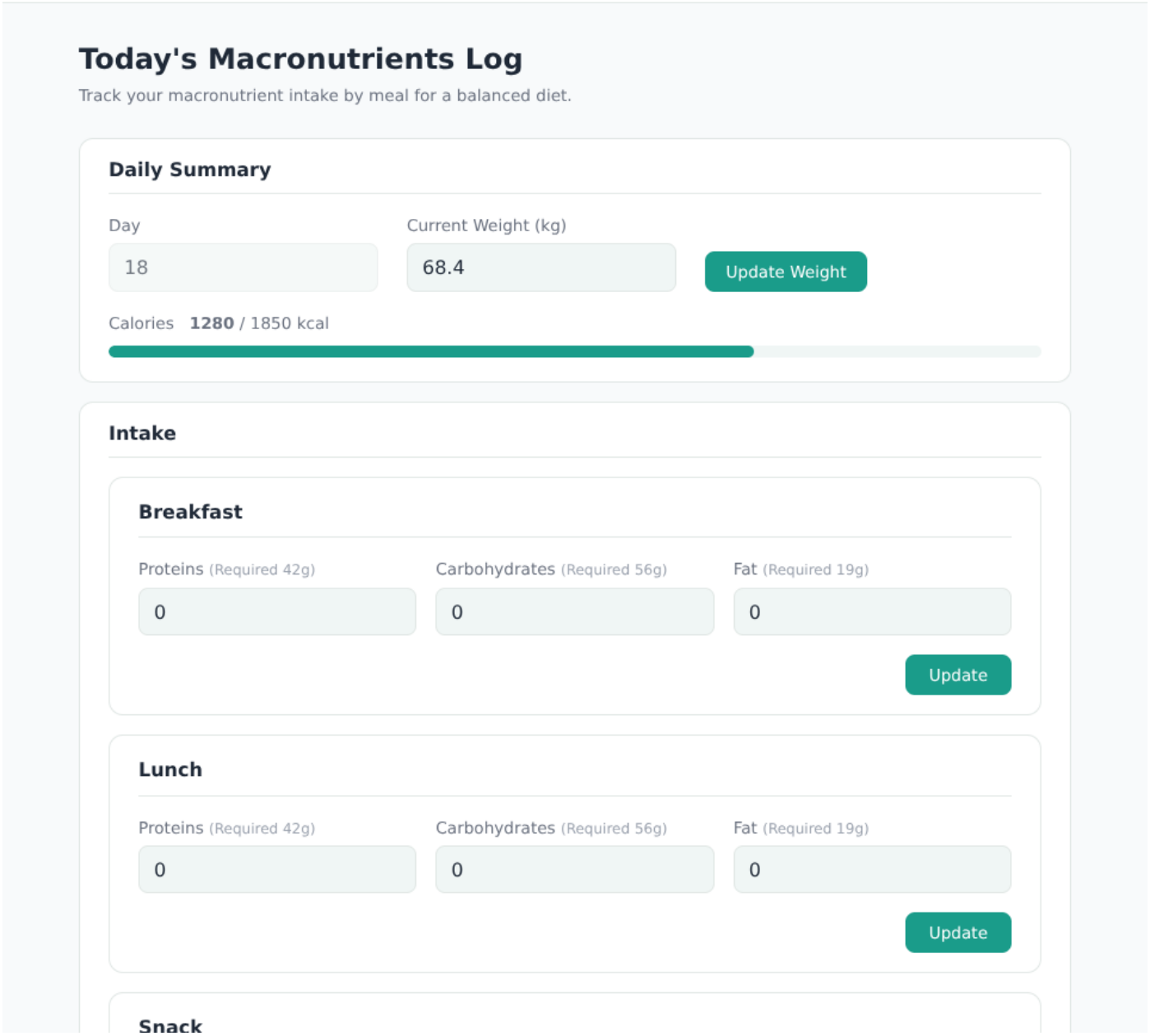
Screenshot C.3 — Dashboard and weight trend



Captured from the running Angular application at /dashboard. Demonstration-only API fixtures provide synthetic progress values; this is not a live account or production database screenshot.

Capture note: local Angular development build with mocked API responses; displayed values are synthetic and illustrative.

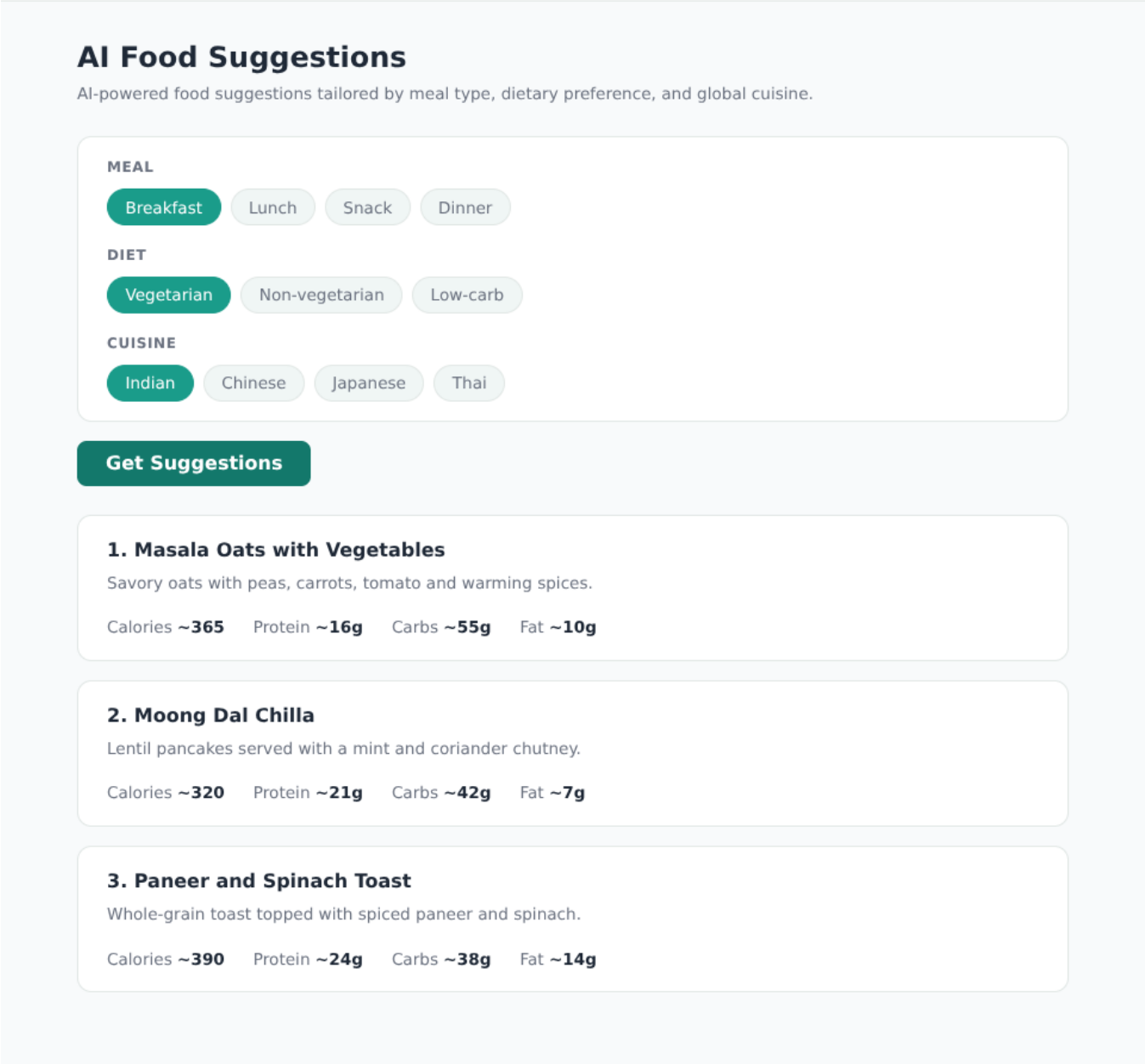
Screenshot C.4 — Daily macronutrient log



Captured from the running Angular application at /progress. The crop shows the daily summary and meal-level macro entry cards. Synthetic demo API values are used; no personal data is present.

Capture note: local Angular development build with mocked API responses; displayed values are synthetic and illustrative.

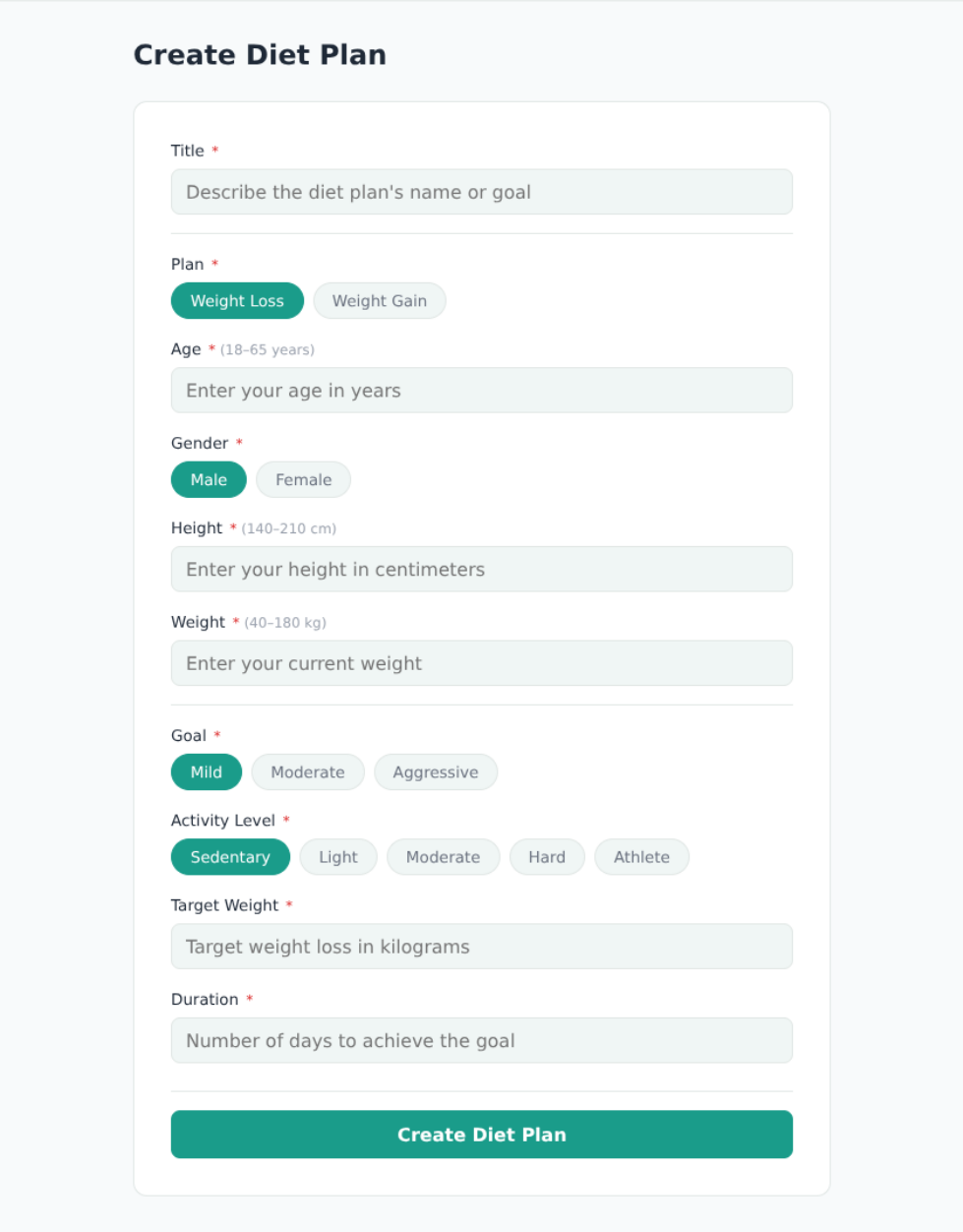
Screenshot C.5 — AI food suggestion filters and results



Captured from the running Angular application at /meals. The UI is live; example suggestion results are mocked for documentation and were not generated by the external model during capture.

Capture note: local Angular development build with mocked API responses; displayed values are synthetic and illustrative.

Screenshot C.6 — Diet-plan creation form



Captured from the running Angular application at /diet-plan-form. The crop shows the complete form in its initial state; no submission was made.